The Ethical Implications of Transforming and Cleaning a Dataset

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In this project, I applied six sequential transformations to the Sports Betting Profiling Dataset to prepare it for analysis. First, I split a single combined-header column into nine distinct fields (bet_id, user_id, bet_type, sport, odds, is_win, stake, gain, and GGR. Second, I standardized the text casing of the bet_type column by converting all entries to lowercase and trimming whitespace. Third, I standardized the string casing in the sport and bet_type columns to find all the unique entries. Fourth, I removed any duplicate rows based on the unique bet_id to guard against double-counting. Finally, fifth and sixth, I computed a new profit column (user_gain – total_stake) and flagged extreme outliers beyond the 99th percentile in stakes or payouts.

Although this is a publicly posted, simulated dataset from Kaggle, sports wagering falls under strict regulatory guidelines (for example, the U.S. Unlawful Internet Gambling Enforcement Act and state gaming commission rules). While this data contains no personal identifiers beyond a synthetic user_id, in a production environment, we would still need to ensure compliance with privacy laws such as GDPR or CCPA and to validate that no personally identifiable information remains in the cleansed data.

Several risks arise from these transformations. Splitting and renaming headers assumes that the delimiter is always correct and that each row contains exactly nine fields, so stray commas or malformed rows could misalign data. Removing duplicates risks discarding legitimate repeat bets if their identifiers were not truly unique. Outlier flags may inadvertently label valid high-stakes wagers as errors. In computing profit, I also assumed that gain equates directly to net return without additional fees or reversals (they certainly do if placed from a valid Sportsbook).

To verify credibility, I sourced the dataset from a reputable public repository and inspected its documentation and sample rows before cleansing. Because it is synthetic, ethical concerns around user privacy are minimal, and the acquisition process requires no private credentials. Nonetheless, transparency is essential when dealing with potentially sensitive data: all transformations are logged in my notebook, and a copy of the raw dataset is preserved.

To mitigate ethical and data-integrity implications, I recommend maintaining an immutable copy of the original data, documenting each transformation in a data dictionary or audit log, and involving domain experts to review outlier thresholds and duplicate-removal logic. Should this workflow be repurposed for real wagering data, strict anonymization and adherence to applicable gaming and privacy regulations would be mandatory to maintain ethical credibility.

References

U.S. Congress. (2006). Unlawful Internet Gambling Enforcement Act of 2006 (Public Law No. 109-347). Retrieved from https://www.congress.gov/109/plaws/publ347/PLAW-109publ347.pdf

European Parliament. (2016). Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 (General Data Protection Regulation). Retrieved from https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32016R0679